

GNIOT Institute of Management Studies – PGDM (Batch 2024-26)**(TRIMESTER –III) END TERM EXAMINATION, MAY -2025****Marketing Analytics****[Time: 2 Hours]****[Maximum Marks: 60]****Following are the course outcomes of the subject: - Marketing Analytics**

CO	Course Outcome (CO)
CO1	Developing a strong ability to analyze financial data, interpret results, and make informed decisions.
CO2	Enhancing problem-solving skills in financial contexts, especially related to project finance, mergers, and feasibility analysis.
CO3	Being able to summarize and present complex financial data and analysis to top management or stakeholders effectively.

*(Note- Faculty can add the more CO's as per their subject.)***SECTION-A****Attempt all parts of the following: (Short Answer Type Questions, do not exceed 150- 200 words)**

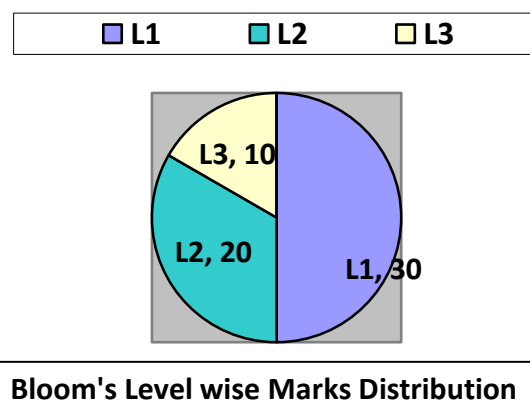
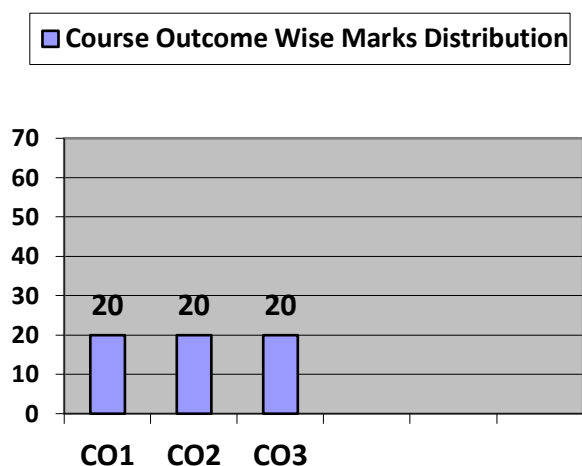
Q.1	Marks	COURSE OUTCOME	BLOOMS LEVEL
(A)	4	CO – 1	BL-1
(B)	4	CO – 1	BL-1
(C)	4	CO – 1	BL-1
(D)	4	CO – 1	BL-1
(E)	4	CO - 1	BL-1

SECTION-B**Attempt ANY TWO questions out of the following: (Long Answer Type Questions, do not exceed 300- 400 words)**

Q. No	Marks	COURSE OUTCOME	BLOOMS LEVEL
2	10	CO – 2	BL-2
3	10	CO – 2	BL-2

SECTION-C**Attempt all parts of the following: (Caselet / Numerical, do not exceed 300- 400 words)**

Q.4	Marks	COURSE OUTCOME	BLOOMS LEVEL
4 (A)	10	CO – 3	BL-1
4 (B)	10	CO – 3	BL-3



Moderated By

IQAC

Dean - Examinations

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GNIOT Institute of Management Studies – PGDM (Batch 2024-26)

(TRIMESTER – III) END TERM EXAMINATION, MAY - 2025

Marketing Analytics

[Time: 2 Hours]

[Maximum Marks: 60]

INSTRUCTIONS:

- (i) There are **THREE** sections in this question paper.
- (ii) Attempt questions from each section as per the directions.
- (iii) Make suitable assumptions wherever necessary.

SECTION-A.

(Short Answer Type Questions, do not exceed 200 words)

Q.1. Attempt all parts of the following:

(5 X 4 = 20)

- (a) Highlight the role of marketing analytics in decision-making? Explain your answer in detail.
- (b) Define various types of variables and where these can be plotted into graph to determine the data points.
- (c) What does CLV stand for in marketing analytics? What is the formula used to calculate CLV?
- (d) What is the significance of Revenue Elasticity is -2? Also, provide the equation for Power Demand curve?
- (e) Define the relationship between price, demand and supply? 'or' what is the relationship between unit price and quantity demanded?

SECTION-B

(Long Answer Type Questions)

Q2. Attempt all questions of the following.

(2 X 10 = 20)

3M is a popular tape used in US for various purposes. The tape can be used for packaging and fixing the things. However, there are a lot of competitors like turtle tapes appeared in the market and capitalized a good market size in recent years. The following graphs and charts are presenting the sales of 3M tape from past two years compiled monthly and yearly in the store. The organization runs an ad campaign and also decides to display the tape on end cap or not. The outputs are presented in figure 2.1, 2.2 Table 2.1 and 2.2. Answer following.

- 1. Infer the sales trend of the tape sales? Provide yearly and monthly comparison as well.
- 2. Explain the impact of advertisement on sales tape? Provide a detailed description. Include highest and lowest selling periods?
- 3. Calculate the average sales count with ads and no ads, and average sales revenue with end cap display and no-end cap display. Provide detailed analysis with average monthly revenue and average tape sales.
- 4. Outline strategy to increase the sales of tape.

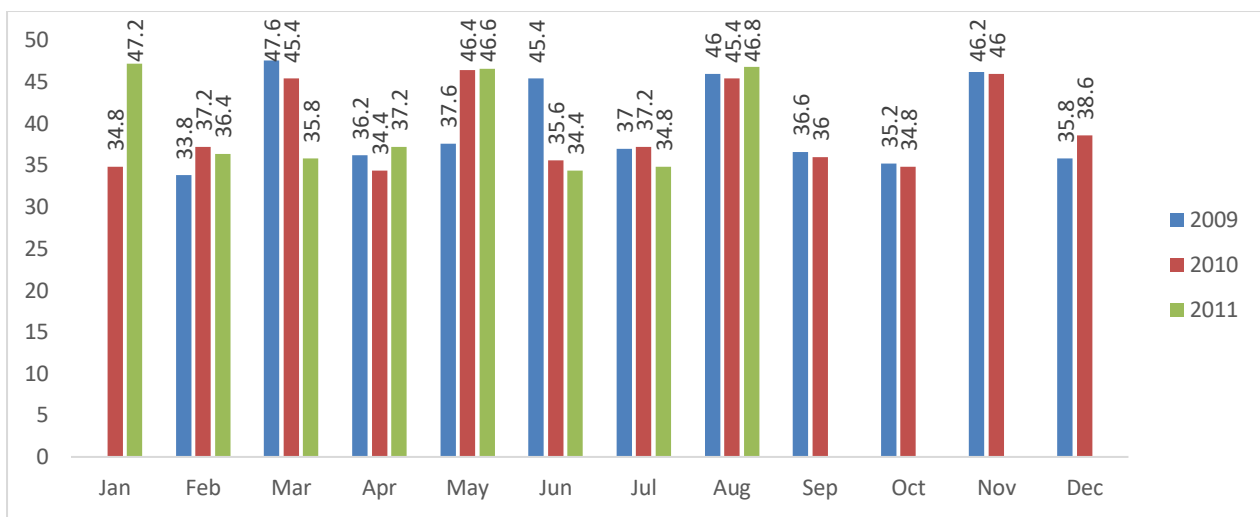


Figure – 2.1.

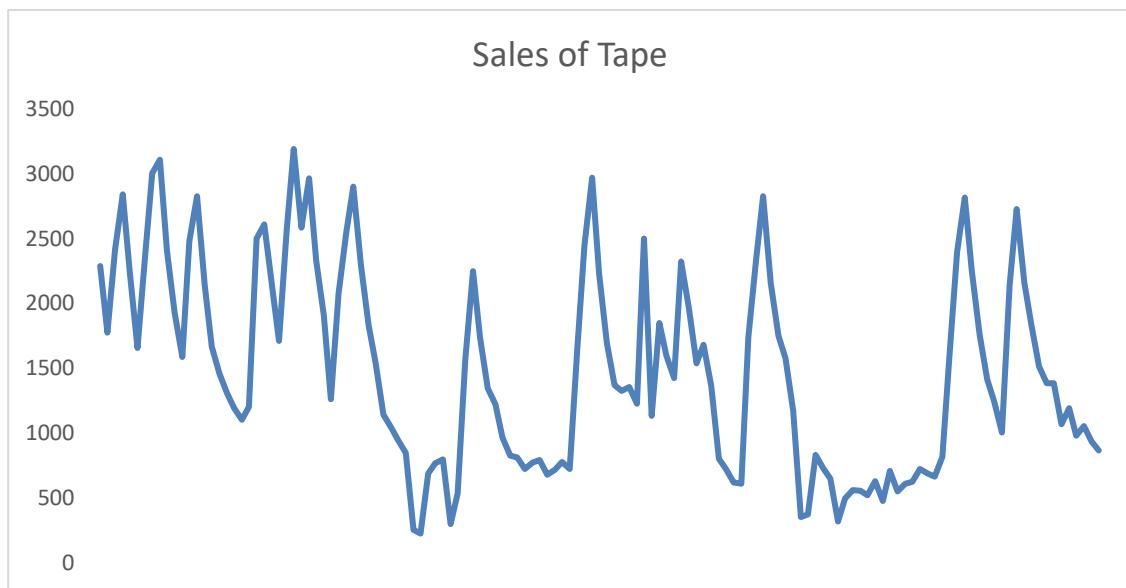


Figure – 2.2

Table – 2.1 (Monthly Sales Data)

Months	Total Count of Units Sold	Total Revenue in USD
Jan	9	82
Feb	12	107.4
Mar	14	128.8
Apr	12	107.8
May	14	130.6
Jun	13	115.4
Jul	12	109
Aug	15	138.2
Sep	8	72.6
Oct	8	70
Nov	10	92.2
Dec	8	74.4
Grand Total	135	1228.4

Table – 2.2 (Impact of Advertising)

No Ad	Ad	Total Revenue	Total Count of Units Sold
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Months	Revenue	Count of Units Sold	Revenue	Count of Units Sold		
Jan	64.4	7	17.6	2	82	9
Feb	81.6	9	25.8	3	107.4	12
Mar	100.4	11	28.4	3	128.8	14
Apr	61.8	7	46	5	107.8	12
May	103.6	11	27	3	130.6	14
Jun	90.4	10	25	3	115.4	13
Jul	80.6	9	28.4	3	109	12
Aug	111	12	27.2	3	138.2	15
Sep	45.8	5	26.8	3	72.6	8
Oct	43.8	5	26.2	3	70	8
Nov	92.2	10			92.2	10
Dec	74.4	8			74.4	8
Grand Total	950	104	278.4	31	1228.4	135

Table – 2.2 (Impact of End Cap)

Revenue Months	Placing tape on End Cap		Grand Total
	No	Yes	
Jan	63.8	18.2	82
Feb	98.8	8.6	107.4
Mar	110.6	18.2	128.8
Apr	99.8	8	107.8
May	121.2	9.4	130.6
Jun	106.4	9	115.4
Jul	109		109
Aug	129.6	8.6	138.2
Sep	72.6		72.6
Oct	70		70
Nov	74	18.2	92.2
Dec	64.6	9.8	74.4
Grand Total	1120.4	108	1228.4

Q3. The graph shows the monthly returns for the standard & Poor stock (S and P return) index and for Dell stock. The beta of stocks is defined as the slope of the least squares line used to predict the monthly return for a stock from the monthly return of the market. Use figure – 3.1 to perform following

1. Summarize the impact of S and P return on dell return? Explain the dependent and independent variables in the problem. Explain the relationship between S and P return and Dell return.
2. Complete the following table using regression equation as mentioned on the graph.

S and P return	Actual Dell return	Predicted Dell Return	Residual Error
0.042	0.282		
0.027	0.158		
-0.014	-0.084		
0.018	-0.080		
0.072	-0.050		
-0.025	-0.172		

0.024	-0.095		
0.052	0.105		
0.040	0.079		
-0.005	-0.073		
0.090	-0.143		
0.019	0.037		
-0.004	0.018		
-0.023	-0.158		
0.021	-0.042		
0.024	-0.043		
-0.067	-0.159		
0.097	NA		NA

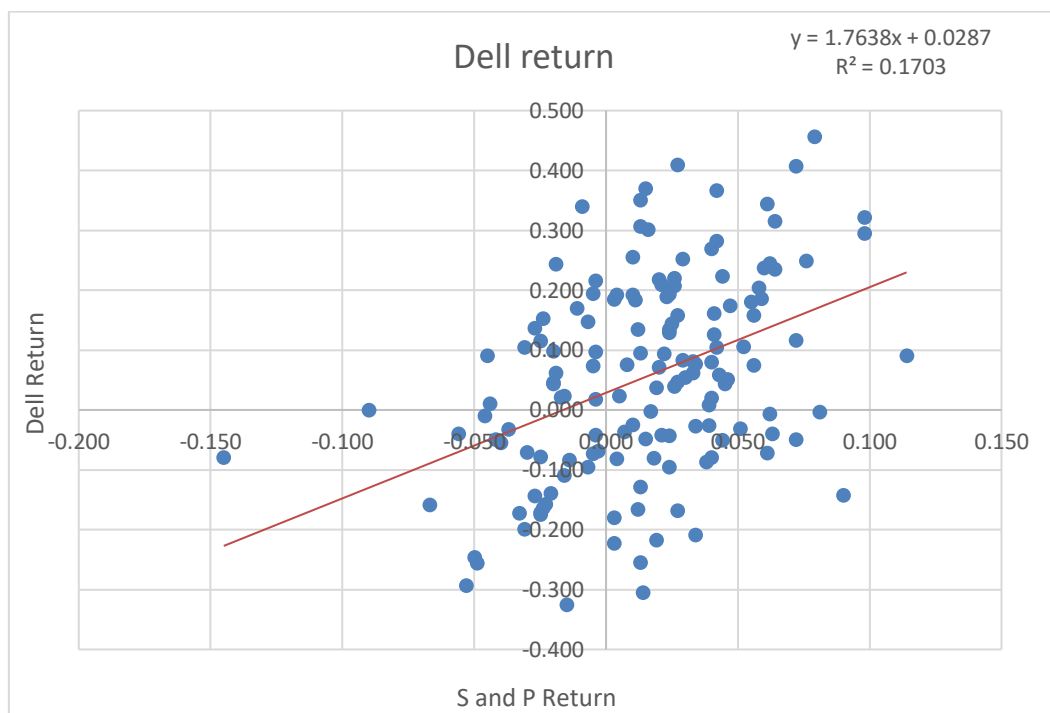


Figure – 3.1

SECTION-C

Q.4. Attempt all parts of the following:

(2 X 10 = 20)

The fuel economy of a car plays an important role. The fuel economy of the car is also dependent on number of Cylinders, displacement, horsepower, weight, and acceleration. Table 4.1 to 4.4 study the relationship between miles per gallon (mpg) and the independent variables mentioned in the problem. Answer the following;

Q4 (A)

- What is the impact of all independent variables collectively on miles per gallon in both cases? Which independent variable(s) has(ve) the highest and lowest impact on miles per gallon? What does the negative beta value(s) of the variable(s) mean in the regression equation?
- If there is a car with 4 cylinders, having a displacement of 230, run on 140 horsepower, and weights 3357 pounds with an acceleration of 9, what would be its miles per gallon average in both the cases?

Q4 (B)

- equations (a) without dropping the non-significant variables (b) after dropping the non-significant variable.
- Complete the following table

Cylinders	Displacement	Horse power	Weight	Acceleration	Actual Mpg	Predicted Mpg (using 4.2) Case 1	Predicted mpg (using 4.3) Case 2	Error (case 1)	Error (Case 2)
8	429	198	4341	10	15				
8	304	120	3962	13.9	15.5				
8	351	142	4054	14.3	15.5				
8	318	145	4140	13.7	15.5				
8	350	170	4165	11.4	15.5				
8	400	190	4325	12.2	15.5				
6	250	100	3278	18	16				
6	250	100	3781	17	16				
6	225	105	3439	15.5	16				
6	250	105	3897	18.5	16				
6	258	110	3632	18	16				
8	302	140	4141	14	16				
8	351	149	4335	14.5	16				
8	304	150	3433	12	16				
8	318	150	4498	14.5	16				
8	318	150	4190	13	16				

Table 4.1 (regression statistics)

Regression Statistics		
Multiple R		0.841
R Square		0.708
Adjusted R Square		0.704
Standard Error		4.247
Observations		392

Table 4.2 (Regression coefficients)

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	46.264	2.669	17.33	0.000	41.016	51.513	41.016	51.513
Cylinders	-0.398	0.411	-0.969	0.333	-1.205	0.409	-1.205	0.409
Displacement	0.000	0.009	-0.009	0.993	-0.018	0.018	-0.018	0.018
Horse Power	-0.045	0.017	-2.716	0.007	-0.078	-0.012	-0.078	-0.012
Weight	-0.005	0.001	-6.351	0.000	-0.007	-0.004	-0.007	-0.004
Acceleration	-0.029	0.126	-0.231	0.817	-0.276	0.218	-0.276	0.218

Table 4.3 (regression statistics with significant independent variables)

Regression Statistics		
Multiple R		0.840
R Square		0.706
Adjusted R Square		0.705
Standard Error		4.240
Observations		392

Table 4.4 (regression coefficients with significant independent variables)

	Coefficient	Standard Error	t Stat	P-value	Lower 95%	Upper 95%	Lower 95.0%	Upper 95.0%
Intercept	45.640	0.793	57.540	0.000	44.081	47.200	44.081	47.200
Horse Power	-0.047	0.011	-4.267	0.000	-0.069	-0.026	-0.069	-0.026
Weight	-0.006	0.001	-11.535	0.000	-0.007	-0.005	-0.007	-0.005